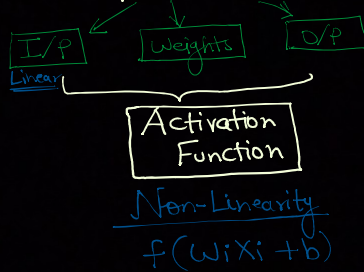


Deep Learning

* Neural Networks

↳ Layers

↳ Neurons

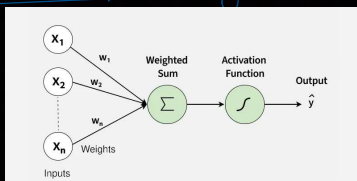


Types of Neural N/w

1. Artificial Neural Network (ANN)
 Complex ML Tasks, All-in-all types
 2. Convolutional Neural Network (CNN)
 Image-based-data, Hold the structure of data
 3. Recurrent Neural Network (RNN)
 - Time Series Analysis (TSA)
 - Natural Language Processing (NLP)
- ↳ LSTM
 ↳ Sequential Modelling (Encoder-Decoder)
 ↳ AutoEncoder
 ↳ Transformer
 ↳ Generative AI

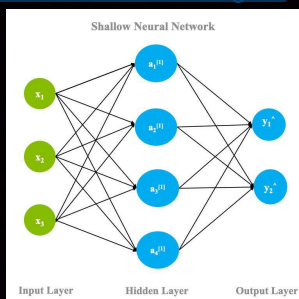
ANN

1. One-neuron/One-layer → Perceptron

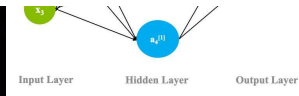


↳ Binary Classification
 0 1

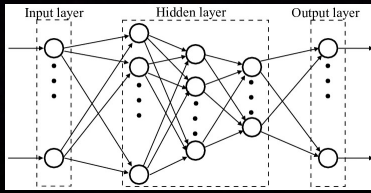
2. Multiple Neuron/One-Layer → Shallow



↳ Hidden Layers = 1
 Requires @ "Activation" Function

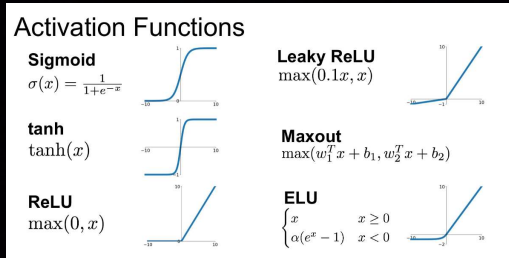


3. Multiple Neurons/Multiple Layers → Deep Neural NW



→ More than 1 hidden layers
→ Multi-Class Classification

Activation Functions



Hidden Layers → ReLU, tanh, Step-wise (Perceptron)

Output Layers → Sigmoid (Binary)
Softmax (Multi-class)

Parameters

→ Layers
 → No neurons in I/P Layer.
 → Contains weight & bias on Hidden.

No. of Params = No. of Neurons X Weights
 + Bias (No. of Neurons)
 + Optimizer Param (No. of Layers)

No. of Parameters $\propto$ Size of the Model

Loss/Cost Function

Task	Last-layer activation	Loss function	Metrics
Binary classification	Sigmoid	Binary crossentropy	Binary accuracy, ROC AUC
Multiclass, single-label classification	Softmax	Categorical crossentropy	Categorical accuracy, top-k categorical accuracy, ROC AUC
Multiclass, multi-label classification	Sigmoid	Binary crossentropy	Binary accuracy, ROC AUC
Regression	None	Mean squared error	Mean absolute error

Table 6.1: Which loss, last-layer activation, and metrics to use for different tasks

Regression	None	Mean absolute error

Table 6.1: Which loss, last-layer activation, and metrics to use for different tasks